

An Unconditional Proof of the Logarithmic Morris Constant-Term Identity

Finite Laurent Coefficients, Augmented Pfaffians, and Compactly Supported Moments

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Abstract

Let $n = 2m + 1$ and $K = 2k + 1$, where a, b, k, m are nonnegative integers. We give an unconditional proof of the logarithmic Morris constant-term identity. The proof uses neither the complex Morris constant-term conjecture nor the evaluation of the Dyson integral for arbitrary complex parameters.

We first define the constant term involving the formal logarithmic product as a linear functional on the finite support of a Laurent polynomial. We then compute the Fourier coefficients of the one-variable principal logarithm and justify the boundary limit by an explicit uniform L^1 majorant. The central evaluation of the alternating logarithmic sum uses the Pfaffian of an augmented skew-symmetric matrix: a congruence transformation of determinant 1 eliminates all angular variables and avoids an informal combinatorial involution. Circularization is performed pointwise only on the full-measure open set of pairwise distinct coordinates, with integrability proved separately.

In the Dyson specialization $a = b = 0$, Good's finite induction gives all integer moments of the circular discriminant. We construct a product of mutually independent Beta random variables with the same moments. After pushing both distributions forward to a common compact interval, the Weierstrass approximation theorem yields moment determinacy, allowing the integer moment formula to be used at the required half-integer exponent. Finally, Laurent Euler derivatives, a finite weight-shift identity for the logarithmic weight, and a three-variable permutation invariance after omitting one logarithm produce a recurrence in the endpoint parameters. A finite Chu–Vandermonde identity and symmetry in a, b complete the proof.

The final section records the interface-level correspondence between the mathematical objects used here and their Lean definitions. The project was built under `leanprover/lean4: v4.28.0`; the final theorem contains no `sorry`, and its axiom report contains only three standard logical dependencies of Lean/Mathlib.

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1 Main Theorem, Conventions, and the Rank-One Case

Let $X = (x_1, \dots, x_n)$ and set

$$\Delta(X) = \prod_{1 \leq i < j \leq n} (x_i - x_j).$$

Throughout the paper,

$$n = 2m + 1, \quad K = 2k + 1, \quad a, b, k, m \in \mathbb{Z}_{\geq 0}.$$

We use the double-factorial conventions

$$0!! = 1!! = 1, \quad (N + 2)!! = (N + 2)N!!.$$

Define the finite Laurent polynomial

$$\Phi_{a,b,k}(X) = \Delta(X) \prod_{i=1}^n x_i^{-m} (1 - x_i)^a (1 - x_i^{-1})^b \prod_{\substack{1 \leq i, j \leq n \\ i \neq j}} \left(1 - \frac{x_i}{x_j}\right)^k. \quad (1)$$

Theorem 1.1 (Logarithmic Morris constant-term identity). *For the parameters above,*

$$\begin{aligned} \text{CT} \left[\Phi_{a,b,k}(X) \prod_{q=1}^m \log \left(1 - \frac{x_{2q}}{x_{2q-1}} \right) \right] \\ = \frac{1}{n!!} \prod_{i=0}^{n-1} \frac{(2a + 2b + iK)!! ((i + 1)K)!!}{(2a + iK)!! (2b + iK)!! K!!}. \end{aligned} \quad (2)$$

Relation to the original theorem. Theorem 1.1 is precisely the assertion of [3, Theorem 1.3] after writing $n = 2m + 1$ and $K = 2k + 1$. In contrast to that theorem’s original derivation, the present proof does not assume the complex Morris constant-term identity [3, (4.5)].

Remark 1.2 (Meaning of the formal logarithm). On the left-hand side of (2), the logarithm is understood as

$$\log(1 - z) = - \sum_{r \geq 1} \frac{z^r}{r}.$$

This does not mean that one first chooses a multivariate formal power series ring and then performs an infinite summation. Since $\Phi_{a,b,k}$ has finite Laurent support, only finitely many exponents can contribute to the constant term. Section 2 defines it directly as a finite sum.

The rank-one case $n = 1$. Here $m = 0$, so there are no logarithmic, Vandermonde, or interaction factors. Hence the left-hand side is

$$\text{CT}[(1 - x)^a(1 - x^{-1})^b] = \sum_{r \geq 0} \binom{a}{r} \binom{b}{r} = \binom{a+b}{a},$$

where the last equality is the finite Vandermonde convolution. The right-hand side has only the term $i = 0$ and equals

$$\frac{(2a+2b)!!}{(2a)!!(2b)!!} = \frac{2^{a+b}(a+b)!}{2^a a! 2^b b!} = \binom{a+b}{a}.$$

Thus the theorem holds for $n = 1$. Unless explicitly stated otherwise, we assume henceforth that $m \geq 1$, and therefore $n \geq 3$. $\square$

2 The Finite Logarithmic Constant-Term Functional

Let

$$F(X) = \sum_{\alpha \in \mathbb{Z}^n} c_\alpha X^\alpha, \quad X^\alpha = \prod_{i=1}^n x_i^{\alpha_i},$$

where only finitely many c_α are nonzero. Define

$$w(\alpha) = \begin{cases} \prod_{q=1}^m \frac{-1}{\alpha_{2q-1}}, & \alpha_{2q-1} > 0, \quad \alpha_{2q} = -\alpha_{2q-1} \quad (1 \leq q \leq m), \\ & \alpha_n = 0, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

and set

$$\Lambda(F) = \sum_{\alpha \in \text{supp}(F)} c_\alpha w(\alpha). \quad (4)$$

Lemma 2.1. *For every finite Laurent polynomial F ,*

$$\Lambda(F) = \text{CT} \left[F(X) \prod_{q=1}^m \log \left(1 - \frac{x_{2q}}{x_{2q-1}} \right) \right].$$

Proof. The q th logarithm contributes precisely the monomials

$$-\frac{x_{2q}^{r_q} x_{2q-1}^{-r_q}}{r_q}, \quad r_q > 0.$$

Its product with X^α has exponent zero in coordinates $2q-1, 2q$ if and only if

$$\alpha_{2q-1} = r_q > 0, \quad \alpha_{2q} = -r_q.$$

The unpaired coordinate requires $\alpha_n = 0$. Under these conditions the coefficient is exactly $\prod_q (-1/\alpha_{2q-1})$; otherwise the contribution is zero. Because $\text{supp}(F)$ is finite, each α corresponds to at most one tuple $(r_1, \dots, r_m)$. Thus the formal constant term is precisely the finite sum (4). $\square$

We henceforth write

$$L(a, b) := \Lambda(\Phi_{a,b,k}).$$

3 The Fourier Bridge and Finite Coefficient Extraction

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with normalized Haar measure dt , and write

$$z(t) = e^{2\pi it}.$$

Let Log denote the principal complex logarithm. To obtain an everywhere-defined measurable function, put

$$\text{Log}_o(\zeta) = \begin{cases} \text{Log}(\zeta), & \zeta \neq 0, \\ 0, & \zeta = 0. \end{cases}$$

The value at zero does not change any integral; it also agrees with the total-function convention used by Lean's `Complex.log`. Set

$$\ell(t) = \text{Log}_o(1 - z(t)).$$

Lemma 3.1 (L^1 integrability and Fourier coefficients). *The function ℓ belongs to $L^1(\mathbb{T})$ and, for every $q \in \mathbb{Z}$,*

$$\int_{\mathbb{T}} z(t)^q \ell(t) dt = \begin{cases} 1/q, & q < 0, \\ 0, & q \geq 0. \end{cases} \quad (5)$$

Proof. We first prove integrability. For $t \neq 0$,

$$|1 - e^{2\pi it}| = 2|\sin(\pi t)|.$$

In a representative neighborhood of $t = 0$, the quantities $|\sin(\pi t)|$ and $|t|$ are comparable. Consequently,

$$|\ell(t)| \leq C + |\log |t||.$$

Since $\int_0^\varepsilon |\log t| dt < \infty$, while ℓ is continuous and bounded away from 0, this proves $\ell \in L^1(\mathbb{T})$.

For $0 \leq r < 1$, let

$$\ell_r(t) = \text{Log}(1 - rz(t)).$$

The absolutely convergent expansion inside the unit disk gives

$$\ell_r(t) = - \sum_{j \geq 1} \frac{r^j z(t)^j}{j}.$$

Termwise integration and character orthogonality yield

$$\int_{\mathbb{T}} z(t)^q \ell_r(t) dt = \begin{cases} -r^{-q}/(-q), & q < 0, \\ 0, & q \geq 0. \end{cases} \quad (6)$$

It remains to justify the limit $r \uparrow 1$. Fix $t \neq 0$ and set

$$d = |1 - z(t)|, \quad u = |1 - rz(t)|.$$

The reverse triangle inequality gives $1 - r \leq u$, and

$$d \leq |1 - rz(t)| + |(r - 1)z(t)| \leq 2u, \quad u \leq 1 + r \leq 2.$$

Hence

$$|\log u| \leq \log 2 + |\log d|.$$

The absolute value of the principal argument is at most π , so

$$|\ell_r(t)| \leq |\log u| + \pi \leq \log 2 + |\ell(t)| + \pi. \quad (7)$$

The right-hand side belongs to $L^1(\mathbb{T})$ and is independent of r . Moreover, $\ell_r(t) \rightarrow \ell(t)$ away from zero. Dominated convergence together with (6) now gives (5). $\square$

Lemma 3.2 (Two-variable Fourier coefficient). *For $p, q \in \mathbb{Z}$,*

$$\int_{\mathbb{T}^2} z(u)^p z(v)^q \text{Log}_o \left(1 - \frac{z(v)}{z(u)} \right) du dv = \begin{cases} -1/p, & p > 0, q = -p, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Proof. Apply the Haar-measure-preserving shear $(u, s) \mapsto (u, u + s)$. The integrand becomes

$$z(u)^{p+q} z(s)^q \ell(s).$$

Fubini's theorem separates the two variables. The first integral forces $p + q = 0$, while the second is evaluated by Lemma 3.1. When $q = -p$, it is nonzero precisely for $p > 0$, and its value is $1/q = -1/p$. $\square$

Proposition 3.3 (Finite-coefficient circular-integral bridge). *Define*

$$\mathcal{L}(t) = \prod_{q=1}^m \text{Log}_o \left(1 - \frac{z(t_{2q})}{z(t_{2q-1})} \right).$$

Then, for every finite Laurent polynomial F ,

$$\Lambda(F) = \int_{\mathbb{T}^n} F(z(t_1), \dots, z(t_n)) \mathcal{L}(t) dt. \quad (9)$$

Proof. The pairs of variables are disjoint. Put

$$f_q(t_{2q-1}, t_{2q}) = \text{Log}_o \left(1 - \frac{z(t_{2q})}{z(t_{2q-1})} \right).$$

Lemma 3.1 and the measure-preserving shear show that each $f_q \in L^1(\mathbb{T}^2)$. By Tonelli's theorem and the disjointness of the variable blocks,

$$\int_{\mathbb{T}^{2m}} \prod_{q=1}^m |f_q| = \prod_{q=1}^m \int_{\mathbb{T}^2} |f_q| < \infty.$$

The unpaired coordinate contributes the factor 1, so $\mathcal{L} \in L^1(\mathbb{T}^n)$; Fubini's theorem may therefore be used to separate the variable blocks. Since a Laurent polynomial is bounded on the compact torus, the integrand in (9) is integrable.

First take $F = X^\alpha$. Apply (8) to each pair and character orthogonality to the unpaired variable. The integral is nonzero exactly under the admissibility conditions in (3), and its value is $w(\alpha)$. The general case follows by finite linearity. $\square$

4 Augmented Pfaffians and Circularization

For an arbitrary $n \times n$ matrix $A = (A_{ij})$, define

$$\text{Alt}_m(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{q=1}^m A_{\sigma(2q-1), \sigma(2q)}. \quad (10)$$

For $c, y_1, \dots, y_n \in \mathbb{C}$, define the skew-symmetric matrix

$$S_{ij}(c, y) = \begin{cases} y_j - y_i + c, & i < j, \\ -(y_i - y_j + c), & j < i, \\ 0, & i = j. \end{cases} \quad (11)$$

Lemma 4.1 (Augmented sawtooth Pfaffian). *If $n = 2m + 1$, then*

$$\text{Alt}_m(S(c, y)) = 2^m m! c^m. \quad (12)$$

Proof. Introduce the index $N = n + 1 = 2m + 2$ and define the $N \times N$ skew-symmetric matrix B by

$$B_{ij} = S_{ij}(c, y) \quad (1 \leq i, j \leq n), \quad B_{iN} = 1, \quad B_{Ni} = -1.$$

Every Pfaffian principal submatrix below is ordered by the inherited natural order of its indices, and we set $\text{Pf}(\emptyset) = 1$. We use the standard convention

$$\text{Pf}(B) = \frac{1}{2^{m+1}(m+1)!} \sum_{\tau \in S_N} \text{sgn}(\tau) \prod_{r=1}^{m+1} B_{\tau(2r-1), \tau(2r)}.$$

Group (10) according to the unpaired vertex $u = \sigma(n)$. Moving u from its u th position in the natural order to the last position requires $n - u$ adjacent transpositions. Since n is odd,

$$(-1)^{n-u} = (-1)^{u+1}.$$

For fixed u , the sum over oriented, ordered pairings of the remaining $2m$ vertices is $2^m m! \text{Pf}(B_{\widehat{u}, \widehat{N}})$, where $B_{\widehat{u}, \widehat{N}}$ denotes the principal submatrix obtained by deleting rows and columns u, N . Therefore

$$\text{Alt}_m(S) = 2^m m! \sum_{u=1}^n (-1)^{u+1} \text{Pf}(B_{\widehat{u}, \widehat{N}}).$$

On the other hand, expansion of the augmented matrix B along its last row and column, using $B_{uN} = 1$, gives

$$\text{Pf}(B) = \sum_{u=1}^n (-1)^{u+1} \text{Pf}(B_{\widehat{u}, \widehat{N}}).$$

The signs agree term by term, so

$$\text{Alt}_m(S) = 2^m m! \text{Pf}(B). \quad (13)$$

Regard B as its associated alternating bilinear form. Let $e_1, \dots, e_N$ be the standard basis and make the change of basis

$$f_i = e_i - y_i e_N \quad (1 \leq i \leq n), \quad f_N = e_N.$$

Its transition matrix is unitriangular and has determinant 1. The Pfaffian congruence formula

$$\text{Pf}(P^T B P) = \det(P) \text{Pf}(B)$$

therefore leaves the Pfaffian unchanged. Standard conventions for Pfaffian expansion and congruence may be found in [6]. For $i < j \leq n$, a direct calculation gives

$$\begin{aligned} B(f_i, f_j) &= B(e_i, e_j) - y_j B(e_i, e_N) - y_i B(e_N, e_j) \\ &= (y_j - y_i + c) - y_j + y_i = c, \end{aligned}$$

and $B(f_i, f_N) = 1$. Thus the strictly upper-triangular part of the new matrix C satisfies

$$C_{ij} = c \quad (1 \leq i < j \leq n), \quad C_{iN} = 1.$$

Every perfect matching contains exactly one edge incident to N ; the remaining m edges lie among the first n vertices. Consequently every Pfaffian term contains the factor c^m , and

$$\text{Pf}(C) = c^m \text{Pf}(J_N),$$

where J_N is the skew-symmetric matrix whose strictly upper-triangular entries are all 1.

Finally, we show that $\text{Pf}(J_{2r}) = 1$. For $r = 0$, this is precisely the empty-Pfaffian convention. Expanding along the first row and using the induction hypothesis,

$$\text{Pf}(J_{2r}) = \sum_{j=2}^{2r} (-1)^j \text{Pf}(J_{2r-2}) = \sum_{j=2}^{2r} (-1)^j = 1.$$

Hence $\text{Pf}(B) = \text{Pf}(C) = c^m$. Substitution into (13) proves (12). $\square$

Define the circular kernel

$$\mathcal{C}_{a,b,K}(t) = \prod_{i=1}^n (1 - z(t_i))^a (1 - z(t_i)^{-1})^b \prod_{1 \leq i < j \leq n} |z(t_i) - z(t_j)|^K. \quad (14)$$

When $a \neq b$, it is generally complex-valued; for this reason we call it a circular kernel rather than a density.

Lemma 4.2 (The alternating logarithmic sum in an ordered chamber). *Suppose*

$$0 \leq \theta_1 < \cdots < \theta_n < 2\pi, \quad z_j = e^{i\theta_j},$$

and put

$$A_{ij} = \text{Log}_o(1 - z_j/z_i) \quad (1 \leq i, j \leq n).$$

In particular, the total-function convention of Section 3 gives $A_{ii} = \text{Log}_o(0) = 0$. Then

$$\text{Alt}_m(A) = m!(-\pi i)^m. \quad (15)$$

Proof. For $i < j$, let

$$\delta = \theta_j - \theta_i \in (0, 2\pi).$$

We have

$$\begin{aligned} 1 - e^{i\delta} &= 2 \sin \frac{\delta}{2} e^{i(\delta/2 - \pi/2)}, \\ 1 - e^{-i\delta} &= 2 \sin \frac{\delta}{2} e^{i(\pi/2 - \delta/2)}. \end{aligned}$$

Both displayed arguments lie in $(-\pi, \pi)$, so the principal branch gives

$$\text{Log}(1 - e^{i\delta}) - \text{Log}(1 - e^{-i\delta}) = i(\delta - \pi).$$

Both arguments of the logarithms are nonzero, so Log_o agrees here with the principal logarithm. Therefore

$$A_{ij} - A_{ji} = i(\theta_j - \theta_i - \pi). \quad (16)$$

Independently swapping the two positions in every input pair of (10) gives

$$\text{Alt}_m(A - A^\top) = 2^m \text{Alt}_m(A).$$

By (16),

$$A - A^\top = S(-\pi i, (i\theta_j)_j).$$

Lemma 4.1, followed by cancellation of 2^m , proves (15). $\square$

Lemma 4.3 (The phase in an ordered chamber). *In the ordered chamber of Lemma 4.2,*

$$\Phi_{a,b,k}(z_1, \dots, z_n) = (-i)^{mn} \mathcal{C}_{a,b,K}(t). \quad (17)$$

Proof. For each $i < j$,

$$\left(1 - \frac{x_i}{x_j}\right)^k \left(1 - \frac{x_j}{x_i}\right)^k = (-1)^k \frac{(x_i - x_j)^{2k}}{(x_i x_j)^k}.$$

Hence

$$\Phi_{a,b,k} = \varepsilon \Delta^K \prod_{i=1}^n x_i^{-(b+mK)} (1 - x_i)^{a+b}, \quad \varepsilon = (-1)^{k\binom{n}{2}+bn}. \quad (18)$$

In the ordered chamber, for every pair we use

$$z_i - z_j = (-i)e^{i(\theta_i + \theta_j)/2} |z_i - z_j|.$$

In the midpoint phase of Δ^K , each z_i occurs $n - 1 = 2m$ times, exactly cancelling z_i^{-mK} . Rewriting $(1 - z_i)^{a+b} z_i^{-b}$ as $(-1)^b (1 - z_i)^a (1 - z_i^{-1})^b$, and setting $C = \binom{n}{2}$, we find that the endpoint factors contribute $(-1)^{bn}$ and the pairwise decomposition of Δ^K contributes $(-i)^{KC}$. Together with ε from (18), the total phase is

$$\begin{aligned} \varepsilon (-1)^{bn} (-i)^{KC} &= (-1)^{kC} (-i)^{KC} \\ &= (-i)^{2kC} (-i)^{(2k+1)C} \\ &= (-i)^{(4k+1)C} \\ &= (-i)^C. \end{aligned}$$

Since $C = \binom{n}{2} = mn$, this is $(-i)^{mn}$, proving (17). $\square$

Proposition 4.4 (Circularization formula). *For all $a, b, k, m \in \mathbb{Z}_{\geq 0}$,*

$$L(a, b) = \frac{m! \pi^m}{n!} \int_{\mathbb{T}^n} \mathcal{C}_{a,b,K}(t) dt. \quad (19)$$

Proof. For $\sigma \in S_n$ and $t \in \mathbb{T}^n$, fix the convention

$$(\sigma t)_j := t_{\sigma(j)} \quad (1 \leq j \leq n).$$

By Proposition 3.3, $L(a, b)$ is the integral of $\Phi(z(t)) \mathcal{L}(t)$. The Laurent kernel is bounded on the compact torus, and $\mathcal{L}$ is integrable; hence their product is integrable. Define

$$\mathcal{A}(t) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \mathcal{L}(\sigma t).$$

Changes of variables by permutations preserve Haar measure. Since Φ is alternating, every $\Phi(t) \mathcal{L}(\sigma t)$ is integrable, and the finite alternating sum may be integrated term by term:

$$n! \int_{\mathbb{T}^n} \Phi(z(t)) \mathcal{L}(t) dt = \int_{\mathbb{T}^n} \Phi(z(t)) \mathcal{A}(t) dt. \quad (20)$$

In any ordered chamber, let

$$A_{ij} = \text{Log}_o \left(1 - \frac{z(t_j)}{z(t_i)} \right).$$

Termwise comparison of the definitions of $\mathcal{L}$, $\mathcal{A}$, and (10) gives

$$\mathcal{A}(t) = \text{Alt}_m(A). \quad (21)$$

This is the precise interface between alternation under the integral and Lemma 4.2.

Let $U \subset \mathbb{T}^n$ be the set of tuples with pairwise distinct coordinates. Its complement is a finite union of diagonal sets $\{t_i = t_j\}$, each of Haar measure zero, so U has full measure. For $t \in U$, choose the unique representative $\theta_i \in [0, 2\pi)$ of each $t_i \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$, with $t_i = \theta_i/(2\pi)$, and then the unique sorting permutation ρ for which the representatives of $(\rho t)_1, \dots, (\rho t)_n$ are strictly increasing. By the convention above and by alternation,

$$\Phi(\rho t) = \operatorname{sgn}(\rho)\Phi(t), \quad \mathcal{A}(\rho t) = \operatorname{sgn}(\rho)\mathcal{A}(t),$$

whereas $\mathcal{C}_{a,b,K}(\rho t) = \mathcal{C}_{a,b,K}(t)$. Thus $\Phi\mathcal{A}$ is permutation-invariant, and Lemmas 4.2 and 4.3 apply in the sorted chamber. Since

$$(-i)^{mn}(-i)^m = (-i)^{m(n+1)} = 1$$

and $m(n+1) = 2m(m+1)$ is divisible by 4, we obtain on U

$$\Phi(z(t))\mathcal{A}(t) = m!\pi^m \mathcal{C}_{a,b,K}(t).$$

The function $\mathcal{C}_{a,b,K}$ is Borel measurable, since it is a finite product of continuous functions on $\mathbb{T}^n$. This is an almost-everywhere identity; no meaningless product $0 \cdot \infty$ is asserted at collision points. The left-hand side is integrable, so the right-hand side is integrable as well. Substitution into (20), followed by Proposition 3.3, gives (19). $\square$

Corollary 4.5 (Exchange of endpoint parameters).

$$L(a, b) = L(b, a).$$

Proof. In (19), make the Haar-measure-preserving change of variables $t \mapsto -t$. Then $z_i \mapsto z_i^{-1}$, the absolute Vandermonde factor is unchanged, and the two endpoint factors are interchanged. $\square$

5 An Unconditional Evaluation in the Dyson Specialization

In this section set $a = b = 0$ and define

$$Y(t) = \prod_{1 \leq i < j \leq n} |z(t_i) - z(t_j)|^2. \quad (22)$$

Then $\mathcal{C}_{0,0,K}(t) = Y(t)^{K/2}$. We begin only with the finite Dyson identity at nonnegative integer exponents; no evaluation at general complex parameters is assumed.

5.1 Good's Finite Dyson Identity

For a finite index set I and $\mathbf{a} = (a_i)_{i \in I} \in \mathbb{Z}_{\geq 0}^I$, put

$$R_i(X) = \prod_{\substack{j \in I \\ j \neq i}} \left(1 - \frac{x_i}{x_j}\right), \quad D_{I,\mathbf{a}}(X) = \prod_{i \in I} R_i(X)^{a_i}.$$

Proposition 5.1 (Good–Dyson).

$$\operatorname{CT}[D_{I,\mathbf{a}}] = \frac{(\sum_{i \in I} a_i)!}{\prod_{i \in I} a_i!}. \quad (23)$$

Proof. If $I = \emptyset$, both sides are 1. Assume henceforth that $I \neq \emptyset$ and regard the x_i as algebraically independent. The sum of the Lagrange basis polynomials evaluated at 0 gives, first in the rational function field $\mathbb{Q}(x_i \mid i \in I)$,

$$1 = \sum_{i \in I} \frac{1}{R_i}.$$

Clearing the denominator $\prod_{i \in I} R_i$ turns this into the Laurent polynomial identity

$$\prod_{i \in I} R_i = \sum_{i \in I} \prod_{\substack{j \in I \\ j \neq i}} R_j.$$

If every $a_i > 0$, multiplication by $\prod_i R_i^{a_i-1}$ yields

$$D_{I,\mathbf{a}} = \sum_{i \in I} D_{I,\mathbf{a}-\mathbf{e}_i}. \quad (24)$$

If $a_h = 0$, then

$$D_{I,\mathbf{a}} = D_{I \setminus \{h\}, \mathbf{a}|_{I \setminus \{h\}}} \prod_{i \in I \setminus \{h\}} \left(1 - \frac{x_i}{x_h}\right)^{a_i}.$$

The first factor is independent of x_h , while the second contains only nonpositive powers of x_h and has constant term 1 in x_h . Hence

$$\text{CT}[D_{I,\mathbf{a}}] = \text{CT}[D_{I \setminus \{h\}, \mathbf{a}|_{I \setminus \{h\}}}] \quad (25)$$

The candidate

$$G(I, \mathbf{a}) = \frac{(\sum_i a_i)!}{\prod_i a_i!}$$

satisfies the same recurrence and boundary relation. Indeed, if every $a_i > 0$ and $A = \sum_i a_i$, then

$$\sum_i G(I, \mathbf{a} - \mathbf{e}_i) = G(I, \mathbf{a}) \sum_i \frac{a_i}{A} = G(I, \mathbf{a}).$$

If $a_h = 0$, then $0! = 1$ gives directly

$$G(I, \mathbf{a}) = G(I \setminus \{h\}, \mathbf{a}|_{I \setminus \{h\}}).$$

Strong induction on $\sum_i a_i + |I|$, starting from the empty index set, proves (23). $\square$

Take $I = \{1, \dots, n\}$ and $a_i = q$. On the unit circle,

$$\prod_{i \neq j} \left(1 - \frac{z_i}{z_j}\right)^q = Y(t)^q.$$

The torus integral of a finite Laurent polynomial is its constant term, so

$$\int_{\mathbb{T}^n} Y(t)^q dt = \frac{(nq)!}{(q!)^n} \quad (q \in \mathbb{Z}_{\geq 0}). \quad (26)$$

5.2 A Beta Product and the Gauss Multiplication Formula

For the moment assume $n \geq 2$. Let $B_1, \dots, B_{n-1}$ be mutually independent random variables with

$$B_r \sim \text{Beta}\left(\frac{r}{n}, 1 - \frac{r}{n}\right) \quad (1 \leq r \leq n-1),$$

and set

$$Z = n^n \prod_{r=1}^{n-1} B_r. \quad (27)$$

Lemma 5.2 (Mellin transform of the Beta product). *Assume $n \geq 2$. For every $s \geq 0$,*

$$\mathbb{E}[Z^s] = \frac{\Gamma(1 + ns)}{\Gamma(1 + s)^n}. \quad (28)$$

Proof. The Beta integral, independence, and $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x + y)$ give

$$\begin{aligned} \mathbb{E}[Z^s] &= n^{ns} \prod_{r=1}^{n-1} \frac{B(s + r/n, 1 - r/n)}{B(r/n, 1 - r/n)} \\ &= \frac{n^{ns}}{\Gamma(1 + s)^{n-1}} \prod_{r=1}^{n-1} \frac{\Gamma(s + r/n)}{\Gamma(r/n)}. \end{aligned} \quad (29)$$

For completeness, we do not invoke the Gauss formula as an unexplained black box. For $x > 0$, define

$$G_n(x) = n^{x-1} \frac{\prod_{r=0}^{n-1} \Gamma((x + r)/n)}{\prod_{r=0}^{n-1} \Gamma((1 + r)/n)}.$$

Using $\Gamma(u + 1) = u\Gamma(u)$ and cancelling consecutive factors gives

$$G_n(x + 1) = xG_n(x), \quad G_n(1) = 1.$$

Clearly $G_n(x) > 0$, and

$$\log G_n(x) = (x - 1) \log n + \sum_{r=0}^{n-1} \log \Gamma((x + r)/n) - \text{constant}$$

is convex. The Bohr–Mollerup uniqueness theorem [4, pp. 34–36] therefore yields $G_n(x) = \Gamma(x)$, or equivalently the normalized Gauss formula

$$\frac{\prod_{r=0}^{n-1} \Gamma((x + r)/n)}{\prod_{r=0}^{n-1} \Gamma((1 + r)/n)} = n^{1-x} \Gamma(x). \quad (30)$$

Set $x = 1 + ns$ in (30) and reindex the numerator and denominator products by $r = 1, \dots, n$. Isolating the factor $\Gamma(1 + s)/\Gamma(1)$ corresponding to $r = n$ then gives

$$n^{ns} \prod_{r=1}^{n-1} \frac{\Gamma(s + r/n)}{\Gamma(r/n)} = \frac{\Gamma(1 + ns)}{\Gamma(1 + s)}.$$

Substitution into (29) proves (28). $\square$

5.3 Pushforward Measures and Moment Determinacy on a Compact Interval

Lemma 5.3 (Moment determinacy on a compact interval). *Let μ, ν be probability measures on the same compact interval $[0, C]$. If*

$$\int_0^C x^q d\mu(x) = \int_0^C x^q d\nu(x) \quad (q \in \mathbb{Z}_{\geq 0}),$$

then, for every $f \in C([0, C])$,

$$\int_0^C f d\mu = \int_0^C f d\nu.$$

Proof. Equality of the moments implies by linearity that the integrals of all polynomials agree. Given $f \in C([0, C])$ and $\varepsilon > 0$, the Weierstrass approximation theorem [5] provides a polynomial p such that $\|f - p\|_\infty < \varepsilon$. Since both probability measures have total mass 1,

$$\left| \int (f - p) d\mu \right| \leq \varepsilon, \quad \left| \int (f - p) d\nu \right| \leq \varepsilon.$$

The integrals of p agree; letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Proposition 5.4 (The circular Dyson integral at real parameters). *Assume $n \geq 2$. For $s \geq 0$,*

$$\int_{\mathbb{T}^n} Y(t)^s dt = \frac{\Gamma(1 + ns)}{\Gamma(1 + s)^n}. \quad (31)$$

Proof. The pairwise estimate $|z_i - z_j| \leq 2$ gives

$$0 \leq Y \leq 4^{\binom{n}{2}}.$$

Since $0 \leq B_r \leq 1$, we also have $0 \leq Z \leq n^n$. Put

$$C = 4^{\binom{n}{2}} + n^n.$$

Let $dt^{\otimes n}$ denote the normalized product Haar probability measure on $\mathbb{T}^n$, and let $\mathbb{P}$ be the probability measure governing the mutually independent variables $B_1, \dots, B_{n-1}$. Define on $[0, C]$ the pushforward probability measures

$$\mu = Y_*(dt^{\otimes n}), \quad \nu = Z_*(d\mathbb{P}).$$

For every integer $q \geq 0$, (26) and Lemma 5.2 give

$$\int x^q d\mu = \frac{(nq)!}{(q!)^n} = \int x^q d\nu.$$

For $s > 0$, Lemma 5.3 applies to the continuous function $f(x) = x^s$; for $s = 0$, take $f \equiv 1$. Therefore

$$\int_{\mathbb{T}^n} Y(t)^s dt = \mathbb{E}[Z^s].$$

Equation (28) now proves (31). □

5.4 The Dyson Base Case

Set $s = K/2 = k + 1/2$. Propositions 4.4 and 5.4 give

$$L(0, 0) = \frac{m! \pi^m}{n!} \frac{\Gamma(1 + nK/2)}{\Gamma(1 + K/2)^n}. \quad (32)$$

Since both n and K are odd,

$$\Gamma\left(1 + \frac{r}{2}\right) = \frac{r!! \sqrt{\pi}}{2^{(r+1)/2}} \quad (r \text{ odd}),$$

and hence

$$\frac{\Gamma(1 + nK/2)}{\Gamma(1 + K/2)^n} = \frac{2^m (nK)!!}{\pi^m (K!!)^n}.$$

Moreover, $n! = n!!(2m)!! = n!! 2^m m!$, so

$$L(0, 0) = \frac{1}{n!!} \frac{(nK)!!}{(K!!)^n}. \quad (33)$$

At $a = b = 0$, the right-hand side of Theorem 1.1 telescopes:

$$\frac{1}{n!!} \prod_{i=0}^{n-1} \frac{((i+1)K)!!}{(iK)!! K!!} = \frac{1}{n!!} \frac{(nK)!!}{(K!!)^n}.$$

Thus the base case has been established unconditionally.

6 The Laurent–Euler Recurrence in the Endpoint Parameters

Set

$$E_r(X) = \sum_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=r}} \prod_{j \in T} (-x_j) = (-1)^r e_r(X), \quad (34)$$

and define

$$H_r(a, b) = \Lambda(\Phi_{a,b,k} E_r), \quad 0 \leq r \leq n.$$

In particular, $H_0(a, b) = L(a, b)$. To treat boundary indices uniformly, for $J \subseteq \{1, \dots, n\}$ and $s \in \mathbb{Z}$ define

$$E_s^{(J)} = \sum_{\substack{T \subseteq \{1, \dots, n\} \setminus J \\ |T|=s}} \prod_{j \in T} (-x_j),$$

with the convention that $E_s^{(J)} = 0$ when $s < 0$ or $s > n - |J|$. We abbreviate $E_s^{(i)} = E_s^{\{i\}}$ and $E_s^{(i,j)} = E_s^{\{i,j\}}$. For a fixed i , put

$$P_{r,i} = (1 - x_i) E_r^{(i)}, \quad D_i = x_i \frac{\partial}{\partial x_i}.$$

6.1 A Laurent-Polynomial Identity

Lemma 6.1 (Local shift identity). *If $0 \leq r < n$, then*

$$\begin{aligned} 2 \sum_{i=1}^n D_i(P_{r,i} \Phi_{a,b,k}) &= -(n-r)(2b+rK) \Phi_{a,b,k} E_r \\ &\quad + (r+1)(2a+2+(n-r-1)K) \Phi_{a,b,k} E_{r+1}. \end{aligned} \quad (35)$$

Proof. We first establish the intermediate identities used below. Elementary counting of finite subsets gives

$$\sum_i E_r^{(i)} = (n-r) E_r, \quad \sum_i (-x_i) E_r^{(i)} = (r+1) E_{r+1}. \quad (36)$$

For $i \neq j$, define

$$Q_{r,ij} = (1 - x_i - x_j) E_r^{(i,j)} + x_i x_j E_{r-1}^{(i,j)}.$$

Decomposing subsets according to whether they contain i and j immediately yields

$$x_i P_{r,i} - x_j P_{r,j} = (x_i - x_j) Q_{r,ij}. \quad (37)$$

We next count the multiplicity of each fixed monomial. A given r -element subset occurring in E_r appears in $E_r^{(i,j)}$ $\binom{n-r}{2}$ times. A given $(r+1)$ -element subset occurring in E_{r+1} appears $(r+1)(n-r-1)$ times in $-x_i E_r^{(i,j)} - x_j E_r^{(i,j)}$ and $\binom{r+1}{2}$ times in $x_i x_j E_{r-1}^{(i,j)}$. Therefore

$$\begin{aligned} 2 \sum_{i < j} Q_{r,ij} &= (n-r-1)(n-r) E_r \\ &\quad + (r+1)(2(n-r-1) + r) E_{r+1}. \end{aligned} \quad (38)$$

For integers $d, t, q \geq 0$, set

$$\Psi_{d,t,q} = \Delta^{q+1} \prod_i x_i^{-d} (1 - x_i)^t.$$

In the field of Laurent rational functions,

$$\frac{D_i \Psi_{d,t,q}}{\Psi_{d,t,q}} = -d - \frac{t x_i}{1 - x_i} + (q+1) \sum_{j \neq i} \frac{x_i}{x_i - x_j}.$$

Since $D_i P_{r,i} = -x_i E_r^{(i)}$, the Leibniz rule and (36) show that the endpoint contribution is

$$-2d(n-r)\Psi E_r + 2(r+1)(t+1-d)\Psi E_{r+1}.$$

Combining the Vandermonde contribution in unordered pairs and applying (37), we obtain

$$2(q+1)\Psi \sum_{i<j} \frac{x_i P_{r,i} - x_j P_{r,j}}{x_i - x_j} = 2(q+1)\Psi \sum_{i<j} Q_{r,ij}.$$

Substitution of (38) now gives

$$\begin{aligned} 2 \sum_i D_i(P_{r,i} \Psi_{d,t,q}) &= [-2d(n-r) + (q+1)(n-r-1)(n-r)] \Psi E_r \\ &\quad + [2(r+1)(t+1-d) \\ &\quad + (q+1)(r+1)(2(n-r-1)+r)] \Psi E_{r+1}. \end{aligned} \quad (39)$$

The calculation was performed in the field of Laurent rational functions. Nevertheless, the left-hand side of (39) is a sum of Euler derivatives of Laurent polynomials, while ΨE_r and ΨE_{r+1} on the right-hand side are Laurent polynomials. Thus (39) is also an identity in the Laurent-polynomial ring: all formal denominators $1 - x_i$ and $x_i - x_j$ have disappeared from the final expression.

Finally use the normalized formula (18), that is, substitute

$$d = b + mK, \quad t = a + b, \quad q + 1 = K, \quad n = 2m + 1.$$

The coefficient of E_r simplifies to $-(n-r)(2b+rK)$, and that of E_{r+1} to $(r+1)(2a+2+(n-r-1)K)$. The constant ε is unaffected by Euler differentiation, proving (35). $\square$

6.2 Finite Logarithmic Weight Shifts: All Exponent Branches

Fix the q th logarithmic pair

$$u = 2q - 1, \quad v = 2q,$$

and let $w^{[q]}$ be the weight obtained by deleting this logarithmic pair. Thus it requires

$$\alpha_u = \alpha_v = 0, \quad \alpha_n = 0,$$

each remaining pair must still satisfy (3), and its weight is the product of the remaining $m - 1$ factors. Denote the corresponding finite functional by $\Lambda^{[q]}$.

Lemma 6.2 (Weight-shift identity). *If $\alpha' = \alpha + e_u - e_v$, then, for every $\alpha \in \mathbb{Z}^n$,*

$$\alpha_u w(\alpha) - (\alpha_u + 1)w(\alpha') = w^{[q]}(\alpha). \quad (40)$$

Proof. First suppose that all pairs other than (u, v) and the unpaired coordinate satisfy their respective conditions, and write W for the product of their weights.

If $\alpha_u = p > 0$ and $\alpha_v = -p$, then

$$w(\alpha) = -W/p, \quad w(\alpha') = -W/(p+1),$$

so the left-hand side is $-W + W = 0$; the right-hand side also vanishes because $(\alpha_u, \alpha_v) \neq (0, 0)$. If $(\alpha_u, \alpha_v) = (0, 0)$, then $w(\alpha) = 0$ and $w(\alpha') = -W$, so the left-hand side is W , exactly $w^{[q]}(\alpha)$.

It remains to check every other case. If $\alpha_u < 0$, neither α_u nor $\alpha_u + 1$ is a positive left exponent; in the sole exception that needs to be separated, $\alpha_u = -1$, the second coefficient $\alpha_u + 1$ itself is zero. Thus both terms still vanish. If $\alpha_u = 0$ but $\alpha_v \neq 0$, then α is inadmissible, whereas admissibility of α' would force $\alpha_v - 1 = -1$, hence $\alpha_v = 0$, a contradiction. If $\alpha_u > 0$ but $\alpha_v \neq -\alpha_u$, then α is inadmissible; admissibility of α' would similarly force $\alpha_v - 1 = -(\alpha_u + 1)$, hence $\alpha_v = -\alpha_u$, again a contradiction. Finally, if any other pair is mismatched or $\alpha_n \neq 0$, all three quantities $w(\alpha)$, $w(\alpha')$, and $w^{[q]}(\alpha)$ vanish. This exhausts all exponent branches. $\square$

Corollary 6.3 (Euler boundary formula). *For arbitrary finite Laurent polynomials G and F ,*

$$\Lambda(D_u((x_v - x_u)G)) = \Lambda^{[q]}(x_v G), \quad (41)$$

$$\Lambda(D_v F) = -\Lambda(D_u F). \quad (42)$$

Proof. Write $G = \sum_{\beta} c_{\beta} X^{\beta}$. The contribution of X^{β} to the left-hand side of the first identity is

$$c_{\beta} [\beta_u w(\beta + \mathbf{e}_v) - (\beta_u + 1)w(\beta + \mathbf{e}_u)].$$

Taking $\alpha = \beta + \mathbf{e}_v$ in Lemma 6.2, this is exactly $c_{\beta} w^{[q]}(\beta + \mathbf{e}_v)$, the corresponding contribution on the right. Finite summation proves (41). The second identity follows monomial by monomial: whenever $w(\alpha) \neq 0$, one has $\alpha_v = -\alpha_u$, while inadmissible monomials contribute zero to both sides. $\square$

6.3 Invariance of the Omitted-Log Functional and Boundary Vanishing

Let $w = n$ denote the unpaired index. Because $\Lambda^{[q]}$ requires the exponents of u, v, w all to be zero, it is invariant under every permutation of these three coordinates. For $\tau \in S_n$, we use the convention

$$(F \circ \tau)(x_1, \dots, x_n) := F(x_{\tau(1)}, \dots, x_{\tau(n)}).$$

Lemma 6.4 (Permutation invariance of the omitted-log functional). *If τ permutes only $\{u, v, w\}$, then, for every finite Laurent polynomial F ,*

$$\Lambda^{[q]}(F \circ \tau) = \Lambda^{[q]}(F). \quad (43)$$

Proof. In $F = \sum_{\alpha} c_{\alpha} X^{\alpha}$, reindex the coefficients by $\alpha \mapsto \alpha \circ \tau^{-1}$. Whenever $w^{[q]}$ is nonzero, the three permuted coordinates must all be zero, whereas the other paired coordinates are unchanged. Consequently $w^{[q]}(\alpha \circ \tau^{-1}) = w^{[q]}(\alpha)$, and the finite sum (4) is invariant. $\square$

Decomposition by subsets gives

$$P_{r,u} - P_{r,v} = (x_v - x_u)R_{r,uv}, \quad (44)$$

where

$$R_{0,uv} = 1, \quad R_{r,uv} = E_r^{(u,v)} - E_{r-1}^{(u,v)} \quad (r > 0).$$

Lemma 6.5 (Three-variable boundary vanishing).

$$\Lambda^{[q]}(x_v R_{r,uv} \Phi_{a,b,k}) = 0. \quad (45)$$

Proof. By (18) and the oddness of K , interchanging any two variables changes the sign of $\Phi_{a,b,k}$.

Consider first a subset T in the expansion of $E_d^{(u,v)}$, and the corresponding monomial

$$M_T = x_v \prod_{j \in T} (-x_j) \Phi_{a,b,k}, \quad T \cap \{u, v\} = \emptyset.$$

If $w \notin T$, interchange u and w . Since neither u nor w lies in T , the factor $x_v \prod_{j \in T} (-x_j)$ is fixed; Φ changes sign; and Lemma 6.4 says that the functional is unchanged. Hence

$$\Lambda^{[q]}(M_T) = -\Lambda^{[q]}(M_T),$$

and this value is zero.

If $w \in T$, write

$$M_T = x_v (-x_w) \prod_{j \in T \setminus \{w\}} (-x_j) \Phi.$$

Interchange v and w . The first two factors change from $x_v(-x_w)$ to $x_w(-x_v)$ and hence their product is fixed; the remaining subset avoids v, w ; Φ again changes sign; and the functional is again invariant by Lemma 6.4. This term is therefore also zero. The argument covers every subset in $E_d^{(u,v)}$, including the empty subset when $d = 0$. Since $R_{r,uv}$ is a difference of two such polynomials, (45) follows. $\square$

Proposition 6.6 (Endpoint-shift recurrence). *For $0 \leq r < n$,*

$$(n-r)(2b+rK)H_r(a,b) = (r+1)(2a+2+(n-r-1)K)H_{r+1}(a,b). \quad (46)$$

Proof. Pair $D_u(P_{r,u}\Phi)$ with $D_v(P_{r,v}\Phi)$. By (42), (44), and (41),

$$\begin{aligned} \Lambda(D_u(P_{r,u}\Phi) + D_v(P_{r,v}\Phi)) \\ = \Lambda(D_u((P_{r,u} - P_{r,v})\Phi)) = \Lambda^{[q]}(x_v R_{r,uv}\Phi) = 0, \end{aligned}$$

where the last equality is Lemma 6.5. Sum over all m pairs. For the remaining unpaired coordinate $w = n$, every admissible exponent is zero, so $\Lambda(D_w F) = 0$. Therefore

$$\Lambda\left(\sum_{i=1}^n D_i(P_{r,i}\Phi)\right) = 0. \quad (47)$$

Applying Λ to Lemma 6.1, using (47), and rearranging proves (46). $\square$

7 Chu–Vandermonde Summation and the Main Theorem

The coefficient

$$(r+1)(2a+2+(n-r-1)K)$$

on the right-hand side of (46) is strictly positive. Thus, starting from $H_0(a,b) = L(a,b)$, we solve the recurrence by dividing only by this coefficient and never by $2b+rK$. Induction gives

$$H_r(a,b) = \binom{n}{r} \frac{\prod_{j=0}^{r-1} (2b+jK)}{\prod_{j=0}^{r-1} (2a+2+(n-1-j)K)} L(a,b). \quad (48)$$

Empty products are understood to be 1. In particular, when $b = 0$ and $r = 0$, the recurrence first gives $H_1(a,0) = 0$; induction then gives $H_r(a,0) = 0$ for every $r > 0$. This agrees exactly with the zero factor at $j = 0$ in the numerator of (48), so the boundary case involves no division by zero.

Lemma 7.1 (Finite-step Chu–Vandermonde identity). *If all denominators below are nonzero, then*

$$\sum_{r=0}^n \binom{n}{r} \frac{\prod_{j=0}^{r-1} (B+jK)}{\prod_{j=0}^{r-1} (A+(n-1-j)K)} = \prod_{i=0}^{n-1} \frac{A+B+iK}{A+iK}. \quad (49)$$

Proof. We first prove the denominator-free polynomial identity

$$\sum_{r=0}^n \binom{n}{r} \prod_{j=0}^{r-1} (B+jK) \prod_{j=0}^{n-r-1} (A+jK) = \prod_{i=0}^{n-1} (A+B+iK). \quad (50)$$

For $n = 0$, both sides are 1. In the induction step from n to $n+1$, apply Pascal's identity and shift the index in the second part from r to $r-1$. For each fixed $0 \leq r \leq n$, the two coefficients combine as

$$(A+(n-r)K) + (B+rK) = A+B+nK.$$

The left-hand side is therefore $A + B + nK$ times its counterpart of order n , and the induction hypothesis proves (50). Finally multiply the r th summand of (49) by $\prod_{i=0}^{n-1} (A + iK)$. Cancelling the last r factors produces precisely the r th summand of (50). Division by the nonzero product completes the proof. $\square$

Since

$$\sum_{r=0}^n E_r(X) = \prod_{i=1}^n (1 - x_i),$$

finite linearity gives

$$\sum_{r=0}^n H_r(a, b) = L(a + 1, b). \quad (51)$$

Take

$$A = 2a + 2, \quad B = 2b$$

in Lemma 7.1; all denominators are positive. Substitution of (48) into (51) yields

$$L(a + 1, b) = L(a, b) \prod_{i=0}^{n-1} \frac{2a + 2b + 2 + iK}{2a + 2 + iK}. \quad (52)$$

Let $R(a, b)$ denote the explicit right-hand side of Theorem 1.1. Applying $(N+2)!! = (N+2)N!!$ factor by factor gives

$$R(a + 1, b) = R(a, b) \prod_{i=0}^{n-1} \frac{2a + 2b + 2 + iK}{2a + 2 + iK}. \quad (53)$$

Clearly $R(a, b) = R(b, a)$, while Corollary 4.5 gives $L(a, b) = L(b, a)$.

Proof of Theorem 1.1. Section 5 established $L(0, 0) = R(0, 0)$. First fix $b = 0$. Induction on a using (52) and (53) gives $L(a, 0) = R(a, 0)$. Symmetry in a, b then gives $L(0, b) = R(0, b)$. Finally fix an arbitrary b and induct once more on a . This proves $L(a, b) = R(a, b)$ for all nonnegative integers a, b , which is precisely (2). $\square$

8 Proof Dependencies and Lean Semantic Audit

8.1 Logical Closure

The logical dependencies of the preceding proof are

$$\begin{aligned} \text{Good-Dyson} &\implies \{\mathbb{E}(Y^r) : r \in \mathbb{Z}_{\geq 0}\}, \\ \text{beta product and compact moment determinacy} &\implies \mathbb{E}(Y^{K/2}), \\ \text{Fourier bridge and augmented Pfaffian} &\implies L(0, 0) = R(0, 0), \\ \text{weight shift and three-variable boundary vanishing} &\implies \text{endpoint-shift recurrence}, \\ \text{Chu-Vandermonde and endpoint symmetry} &\implies L(a, b) = R(a, b). \end{aligned}$$

The first implication uses only finite Laurent identities with nonnegative integer exponents. The unique extension from integer to half-integer moments takes place between probability measures supported on the same compact interval; it does not use the complex Morris conjecture.

8.2 Correspondence Between the Mathematical Objects and the Lean Interfaces

The final theorem in the formal development is

`LogarithmicMorrisFull.logarithmicMorris_full.`

Its type proves `LogarithmicMorrisStatement S` for an arbitrary `Setup`. The following table records correspondences at the level of mathematical objects and formal interfaces; it does not assert that every natural-language argument below has been formalized line by line. The explicit definitions in Sections 1 and 2 provide the human-checkable identification of the formal statement with the constant-term identity of [3, Theorem 1.3].

Mathematical object	Concrete Lean interface or theorem
$n = 2m + 1$, $K = 2k + 1$	<code>Setup.odd_rank</code> and <code>Setup.K</code>
The finite Laurent kernel in (1)	<code>morrisKernel</code> ; its coefficient field is $\mathbb{Q}$, and its exponent type is <code>Fin n → Z</code>
The weight (3) and functional (4)	<code>StandardLogAdmissible</code> , <code>standardLogWeight</code> , and <code>standardLogCT</code>
The two sides of the theorem	<code>logarithmicMorrisLHS</code> and <code>logarithmicMorrisRHS</code>
The Fourier bridge and L^1 control	<code>ScratchCircleLogFourier</code> , <code>ScratchPairedLogIntegrable</code> , and <code>ScratchLogCTBridge</code>
The alternating sum of augmented sawtooth values	<code>LogarithmicMorrisPfaffian</code> ; Lean proves the same finite alternating-sum identity by a wholly finite permutation-sum argument, rather than formalizing line by line the augmented-matrix congruence argument in this section
Ordered chambers, arbitrary permutations, and collisions	<code>ScratchLogPfaff</code> , <code>ScratchPointwise</code> , and <code>ScratchPermutationIntegral</code>
Good–Dyson integer moments	<code>ScratchGoodBoundary</code> and <code>LogarithmicMorrisEvenMoments</code>
The beta product and Gauss’s formula	<code>LogarithmicMorrisDysonBetaProduct</code> and <code>AristotleGammaMultiplicationFresh</code>
Pushforward measures and compactly supported moment determinacy	<code>LogarithmicMorrisDysonMomentComparison</code> and <code>AristotleMomentUniquenessFresh</code>
The local Euler identity	<code>ScratchShiftLocal</code>
All branches of the weight-shift identity and three-variable boundary vanishing	<code>ScratchShiftLogZero</code>
The parameter recurrence and final induction	<code>LogarithmicMorrisParameterReduction</code> and <code>LogarithmicMorrisFull</code>

Remark 8.1 (The circular logarithm at collision points). Lean’s `Complex.log` is a total function, with value 0 at 0. The formal files may therefore state identities that hold at collision points as well: the Vandermonde factor makes the Laurent kernel vanish there. In ordinary mathematical usage, the principal logarithm is generally regarded as defined only away from zero. We accordingly used the extension $\text{Log}_\circ(0) = 0$ and performed the analytic calculation in Proposition 4.4 only on the full-measure set of pairwise distinct coordinates. The two conventions give the same integrals and Fourier coefficients.

Remark 8.2 (The rank-one case). When $m = 0$, the logarithmic product over $\text{Fin } m$ is empty, and `StandardLogAdmissible` requires only that the exponent of the unique coordinate be zero. Thus `standardLogCT` genuinely reduces to the ordinary constant term. The file proving the circular Dyson evaluation also treats $n = 1$ as a separate branch, in agreement with the direct calculation in Section 1.

8.3 Reproducible Build and Axiom Report

The complete release repository is publicly available at

<https://github.com/halfblood12312300-cell/logarithmic-morris-lean>.

The Lean proof snapshot audited in this paper is the immutable commit

321df3a4af7231df43bd5b976ac366e44d0a1ce6.

At this audited snapshot, the repository root contains `lakefile.toml`, `lake-manifest.json`, and `lean-toolchain`. The last of these pins

`leanprover/lean4:v4.28.0`,

and that Lake configuration pins Mathlib at `v4.28.0`. On 22 July 2026, the following commands were run from a fresh checkout:

```
git clone https://github.com/halfblood12312300-cell/logarithmic-morris-lean.git
cd logarithmic-morris-lean
git checkout 321df3a4af7231df43bd5b976ac366e44d0a1ce6
lake exe cache get
lake build logarithmicmorris.LogarithmicMorrisFull
lake env lean --trust=0 logarithmicmorris/LogarithmicMorrisAudit.lean
```

The final-target build succeeded (8082 tasks). The final audit command produced the following output:

```
LogarithmicMorrisFull.logarithmicMorris_full
(S : LogarithmicMorrisFull.Setup) :
  LogarithmicMorrisFull.LogarithmicMorrisStatement S
'LogarithmicMorrisFull.logarithmicMorris_full' depends on axioms:
[propxt, Classical.choice, Quot.sound]
```

Thus the axiom closure contains no `sorryAx` and no external axiom introduced for this problem. The three reported axioms are Lean/Mathlib's standard logical dependencies for quotient types, propositional extensionality, and classical choice; none encodes a Morris, Dyson, Selberg, or integral-evaluation assumption. The project release contains exactly the local transitive import closure required by the final theorem, together with its public entry point and audit file. No Lean source file in this release contains a `sorry` declaration. The absence of an unreported logical assumption is not inferred from file names or a textual search: it is certified by the kernel-level `#print axioms` report reproduced above. The repository also includes a GitHub Actions workflow that rebuilds the complete release library, rejects `sorry`, `sorryAx`, and local axiom declarations, uses the pinned Lean `v4.28.0` toolchain, and prints the same axiom closure.

Relation to the Original Conditional Proof

The unconditional route in this paper differs from the derivation in the original article, which was conditional on the complex Morris identity. The conjectural input is replaced precisely in Section 5: Good’s integer Dyson identity determines all moments of a compactly supported random variable, and a beta product then evaluates its $K/2$ moment. The formal verification ensures that all finite algebraic branches, measure pushforwards, integrability arguments, and parameter recurrences are closed. The present paper expands those machine-checked interfaces into a self-contained mathematical proof.

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